

Jason Wakim Richards

+44 77 08 348 077 | jason2richards@gmail.com

github.com/richarj2002 | linkedin.com/in/jason-richards-jwr2002/ | richarj2002.github.io/personal_website/

11 August 2026

Jet Propulsion Laboratory

Multi-Agent Robotics Group

Robotic Mobility & Manipulation Section

Application for Multi-Agent Robotics Fielding Systems Engineer (R5900)

Dear Hiring Committee,

Space exploration has been the defining focus of my career. I have followed JPL's Ingenuity mission closely since its arrival on Mars, and it became the mission that crystallised the direction I wanted my career to take. Ingenuity showed how an aerial robot could extend the reach of planetary exploration, which directly motivated my decision to pursue a PhD in aerial Semantic SLAM. That choice deliberately united my spacecraft and planetary-rover experience with deep, hands-on UAV robotics. The opportunity to help JPL field complex multi-agent aerospace systems for terrestrial and planetary exploration is therefore not simply aligned with my background; it is the goal towards which I have intentionally developed it.

At Airbus Defence & Space, I worked in the AOCS/GNC team on autonomous planetary-rover technology and led LiDAR test activities during Integrated BreadBoarding 3 and 4. These time-constrained campaigns supported an end-to-end Sample Fetch Rover concept demonstration at Technology Readiness Level 6. I designed a networked sensor test bench, integrated LiDAR with rover breadboards using ROS over a local wireless network, and collected and analysed campaign data. I also developed C/C++ software on Linux for an offline GNC simulation that generated synthetic imagery and LiDAR scans. This took my work from early feasibility through representative hardware integration and testing within a fast-paced, multidisciplinary team.

I now develop and deploy autonomous UAV systems in the Autonomous Robotics Group (ARG), in Luxembourg, using ROS 2, PX4, embedded Linux, Raspberry Pi, cameras, IMUs, OpenVINS, Docker, and Kalibr. My work spans Semantic Relational SLAM, localisation, planning, control, simulation, laboratory integration, and flight testing, with the focus on integrating situational awareness into the autonomy pipeline. This includes collaborative missions based on shared semantic maps and multi-robot coordination. Before my PhD, I led the JamSail 3U CubeSat ADCS, developing embedded C software, an EKF for sun-sensor, magnetometer, and IMU fusion, and a C implementation of the IGRF magnetic-field model.

These experiences match the central challenge of this role: moving research autonomy onto real, networked robotic vehicles and making the complete system work reliably under time pressure. I bring practical experience across ROS/ROS 2, C/C++, Python, embedded Linux, Docker, Raspberry Pi, PX4, sensor and payload integration, localisation, planning, control, communications, hardware, and testing. I also bring the systems judgement and soft skills needed to work across disciplines: translating mission needs into requirements and interfaces, communicating test status and integration issues clearly, and troubleshooting across software, sensors, networks, and hardware. My progression from UKSEDS competition volunteer to Competitions Management Co-Lead, involved coordinating four national competitions with volunteers and industrial partners. This demonstrates that I can take ownership, support others, and deliver against fixed milestones.

I would be proud to bring this combination of planetary-rover development, embedded UAV autonomy, spacecraft systems engineering, and collaborative leadership to JPL's Multi-Agent Robotics Group. I am motivated by the work required to turn ambitious robotics research into dependable capability on real vehicles, and I would welcome the opportunity to help the team deploy systems that expand what is possible in terrestrial and planetary exploration.

Yours sincerely,

Jason Wakim Richards